Comparative Analysis of Machine Learning Algorithms for Heart Disease Prediction

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***Abstract*—Heart disease is one of the leading causes of mortality worldwide. Early prediction of heart disease can significantly improve patient outcomes. In this study, a comparative analysis of machine learning algorithms for heart disease prediction is performed using the Cleveland Heart Disease dataset obtained from the UCI Machine Learning Repository. Data preprocessing techniques were applied to handle missing values and convert the target variable into a binary classification problem. Four machine learning algorithms—Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine—were implemented and evaluated. Experimental results indicate that Logistic Regression and Random Forest achieved the highest accuracy of 88.33%. The results demonstrate the effectiveness of machine learning approaches in assisting healthcare professionals in heart disease prediction.**

***Index Terms*—Machine Learning, Heart Disease Prediction, Random Forest, Logistic Regression, Classification.**

I. Introduction

Heart disease is one of the most significant health challenges worldwide. Cardiovascular diseases account for a large number of deaths each year. Early detection and prediction of heart disease are essential for improving patient outcomes.

Machine learning techniques have been widely applied in healthcare to analyze medical data and assist in disease prediction [2], [3]. By analyzing clinical features such as age, blood pressure, cholesterol levels, and heart rate, machine learning models can help identify potential heart disease cases.

In this research, a comparative analysis of different machine learning algorithms for heart disease prediction is conducted. The algorithms evaluated include Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine. The performance of these models is evaluated using accuracy metrics.

II. Related Work

Several studies have explored the use of machine learning techniques for heart disease prediction. Logistic Regression is widely used in medical data analysis because of its simplicity and interpretability. Decision Tree algorithms provide transparent decision rules that help clinicians understand prediction patterns.

Random Forest, an ensemble learning technique, improves prediction accuracy by combining multiple decision trees and reducing overfitting. Previous studies have shown that Random Forest often achieves higher accuracy compared to individual classification models when applied to medical datasets.

Support Vector Machine (SVM) has also been widely used in healthcare applications due to its strong generalization ability and effectiveness in handling high-dimensional data. However, its performance may vary depending on parameter tuning and dataset characteristics.

Recent research indicates that machine learning-based approaches can significantly improve early detection of cardiovascular diseases by analyzing clinical features such as cholesterol level, blood pressure, and chest pain type. These techniques can assist healthcare professionals in making faster and more accurate diagnostic decisions.

Machine learning methods have been widely studied in statistical learning theory and have been applied to various healthcare prediction problems [4].

Several recent studies have demonstrated the effectiveness of machine learning techniques for medical diagnosis. Algorithms such as Random Forest and Support Vector Machine have shown promising results when applied to healthcare datasets. These models are capable of identifying complex patterns within clinical data and assisting physicians in early diagnosis of diseases. As a result, machine learning has become an important tool in modern healthcare analytics.

III. Methodology

The dataset used in this study is the Cleveland Heart Disease dataset obtained from the UCI Machine Learning Repository. The dataset consists of 303 instances and 14 attributes representing clinical features such as age, sex, chest pain type, blood pressure, cholesterol level, and maximum heart rate.

Data preprocessing was performed to handle missing values represented by the symbol "?". These values were replaced with NaN and removed from the dataset. The target variable was converted into a binary classification problem where 0 indicates no heart disease and 1 indicates the presence of heart disease.

The dataset was divided into training and testing sets using an 80–20 split ratio. Four machine learning algorithms were implemented: Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine. Model performance was evaluated using the accuracy metric.

The overall workflow of the heart disease prediction system is illustrated in Fig. 1.

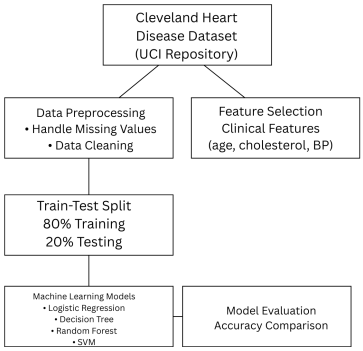

Fig. 1. Workflow of the heart disease prediction system

A. Machine Learning Algorithms

Four machine learning algorithms were used in this study to evaluate their performance in predicting heart disease.

Logistic Regression: Logistic Regression is a statistical classification algorithm commonly used for binary classification problems. It estimates the probability of a binary outcome using a logistic function.

Decision Tree: Decision Tree is a supervised learning algorithm that splits the dataset into branches based on feature values. It is widely used due to its interpretability and simplicity.

Random Forest: Random Forest is an ensemble learning method that constructs multiple decision trees and combines their predictions to improve accuracy and reduce overfitting.

Support Vector Machine: Support Vector Machine (SVM) is a powerful classification algorithm that finds the optimal hyperplane separating data points belonging to different classes.

B. Dataset Description

The Cleveland Heart Disease dataset contains 303 patient records with 14 clinical attributes that are commonly used for heart disease diagnosis. These attributes include age, sex, chest pain type, resting blood pressure, serum cholesterol level, fasting blood sugar, resting electrocardiographic results, maximum heart rate achieved, exercise-induced angina, ST depression, slope of peak exercise ST segment, number of major vessels colored by fluoroscopy, thalassemia, and the target variable indicating the presence of heart disease.

These features represent important clinical indicators that help physicians evaluate cardiovascular conditions. The dataset has been widely used in machine learning research for benchmarking heart disease prediction models.

C. Data Preprocessing

Before training the machine learning models, the dataset was preprocessed to improve data quality. Missing values represented by the symbol "?" were identified and replaced with NaN values. Rows containing missing values were removed to ensure reliable model training.

The target variable was transformed into a binary classification problem where 0 represents the absence of heart disease and 1 represents the presence of heart disease. Feature values were then prepared for training and testing using the Scikit-learn library in Python.

D. Experimental Setup

The machine learning models were implemented using the Python programming language. The Scikit-learn library was used for implementing the classification algorithms and performing model evaluation.

The dataset was divided into training and testing sets using the train$_{t}est_{s}plitfunctionwithan80–20splitratio.LogisticRegression, De$

After training, the models were evaluated on the testing dataset to measure prediction accuracy. The results obtained from each algorithm were then compared to determine which model performed best for heart disease prediction.

The training process involves fitting each machine learning model on the training dataset and then evaluating the trained models using the testing dataset. The purpose of this process is to measure how well the models generalize to unseen data. By comparing the performance of different algorithms, it becomes possible to determine which model is most suitable for heart disease prediction using the given dataset.

E. Evaluation Metric

The performance of the machine learning models was evaluated using the accuracy metric. Accuracy represents the proportion of correctly classified instances among the total number of predictions made by the model.

Accuracy is calculated using the following formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP represents true positives, TN represents true negatives, FP represents false positives, and FN represents false negatives.

Accuracy is widely used for classification problems and provides a straightforward way to compare the performance of different machine learning algorithms.

IV. Results and Discussion

The performance of the machine learning models was evaluated using accuracy. Logistic Regression and Random Forest achieved the highest accuracy of 88.33%, while Decision Tree achieved 81.67%. The Support Vector Machine model achieved the lowest accuracy of 66.67%.

The experimental results show that Logistic Regression and Random Forest achieved the highest prediction accuracy of 88.33%. This indicates that both statistical and ensemble-based models perform well for heart disease prediction using the Cleveland dataset.

Decision Tree achieved an accuracy of 81.67%, which is slightly lower but still demonstrates acceptable performance.

The Support Vector Machine model achieved the lowest accuracy of 66.67%, which may be due to the limited size of the dataset and lack of parameter tuning.

These findings suggest that ensemble methods such as Random Forest can effectively capture complex relationships among clinical features and improve prediction performance. The results also demonstrate that machine learning models can serve as valuable tools in assisting medical professionals in early diagnosis of cardiovascular diseases.

The performance comparison indicates that ensemble learning methods such as Random Forest are highly effective for classification tasks involving medical datasets. Random Forest benefits from combining multiple decision trees, which helps reduce overfitting and improves generalization performance.

Logistic Regression also achieved high accuracy due to the relatively small number of features and the binary nature of the classification problem. However, the Support Vector Machine model produced lower accuracy in this experiment, which may be attributed to the limited dataset size and the absence of extensive hyperparameter tuning.

The comparative analysis highlights the importance of selecting appropriate machine learning algorithms for medical prediction tasks. Ensemble-based methods such as Random Forest are capable of capturing complex patterns in clinical datasets, which leads to improved prediction accuracy. These models can assist healthcare professionals by providing data-driven insights that support early diagnosis and treatment planning.

A. Model Performance Analysis

The comparative evaluation shows that Logistic Regression and Random Forest provide superior performance compared to Decision Tree and Support Vector Machine. Random Forest benefits from ensemble learning, which combines multiple decision trees to improve prediction stability and reduce overfitting. Logistic Regression also performs well due to the relatively simple structure of the dataset and the binary classification task.

Table I presents the accuracy comparison of the evaluated machine learning algorithms. Logistic Regression and Random Forest achieved the highest prediction accuracy among the models.

The graphical comparison of the machine learning algorithms is shown in Fig. 2.

TABLE I
ACCURACY COMPARISON OF MACHINE LEARNING ALGORITHMS

Algorithm	Accuracy (%)
Logistic Regression	88.33
Decision Tree	81.67
Random Forest	88.33
Support Vector Machine	66.67

The performance of the classification model was further evaluated using precision, recall, and F1-score metrics. Precision represents the proportion of correctly predicted positive cases among all predicted positives, while recall measures the ability of the model to identify actual positive cases. The F1-score provides a balance between precision and recall. The results show that the model achieved a precision of 0.88, recall of 0.88, and F1-score of 0.88, indicating strong and balanced classification performance.

TABLE II
PERFORMANCE METRICS OF THE PREDICTION MODEL

Metric	Precision	Recall	F1-Score
Model Performance	0.88	0.88	0.88

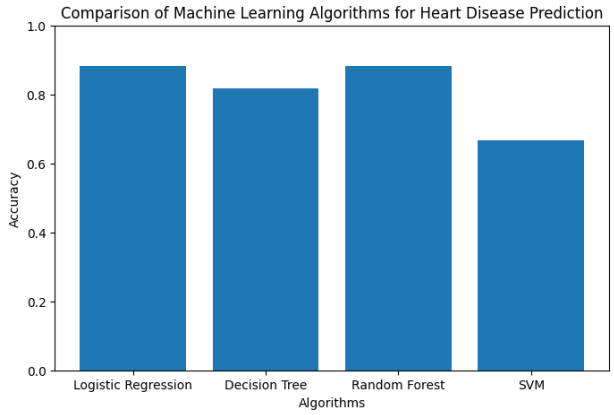

Fig. 2. Comparison of machine learning algorithms for heart disease prediction

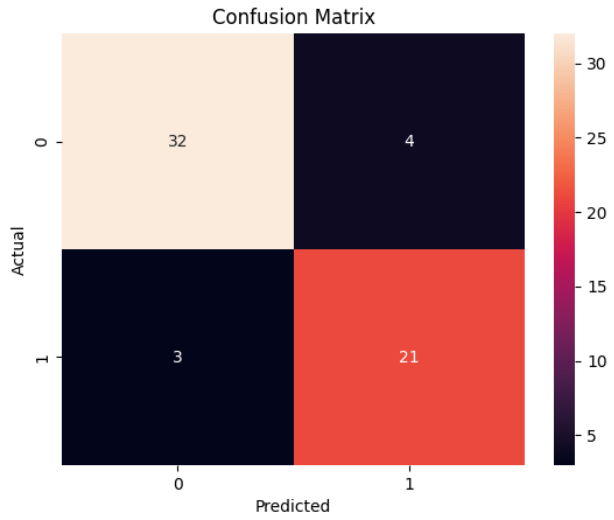

Fig. 3. Confusion matrix for heart disease prediction

The Receiver Operating Characteristic (ROC) curve illustrates the classification performance of the model across different threshold values. The curve represents the relationship between the true positive rate and false positive rate. A larger area under the curve (AUC) indicates better classification performance. The ROC analysis demonstrates that the model is capable of effectively distinguishing between patients with and without heart disease.

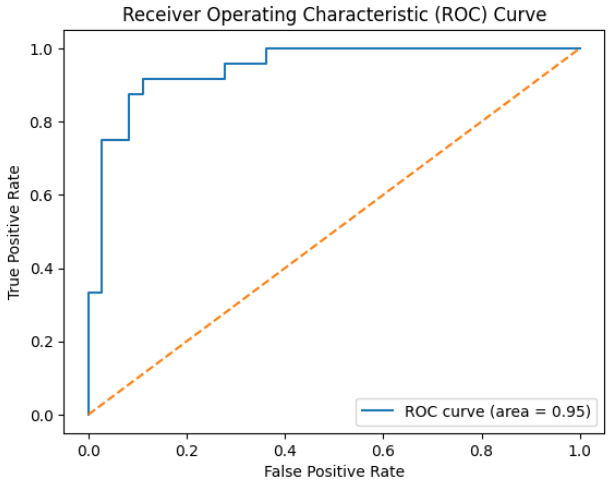

Fig. 4. ROC curve for the heart disease prediction model

B. Limitations

Although the proposed approach demonstrates promising results, the study has some limitations. The Cleveland Heart Disease dataset contains a relatively small number of samples, which may limit the generalization capability of the models. Additionally, only accuracy was used as the evaluation metric. Future studies may incorporate additional evaluation metrics such as precision, recall, and F1-score to provide a more comprehensive performance analysis.

V. Conclusion

This study presented a comparative analysis of machine learning algorithms for heart disease prediction using the Cleveland Heart Disease dataset. Four classification models—Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine—were implemented and evaluated using accuracy as the performance metric.

The experimental results indicate that Logistic Regression and Random Forest achieved the highest prediction accuracy of 88.33%. These findings suggest that both statistical and ensemble-based machine learning approaches are effective for heart disease prediction.

The results demonstrate the potential of machine learning techniques in assisting healthcare professionals in early detection and diagnosis of cardiovascular diseases. Future work may explore deep learning techniques, feature selection methods, and larger medical datasets to further improve prediction accuracy and model robustness.

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